Laeeq Aslam, PhD

Physics-Informed Machine Learning | Renewable Energy Forecasting | Time-Series Modeling | Edge AI

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Professional Summary

PhD in Control Science and Engineering with research focused on physics-informed machine learning, renewable-energy forecasting, spatio-temporal time-series modeling, and edge AI. My doctoral work developed short-term wind-speed prediction models that combine data-driven learning with physical constraints, adaptive feature selection, and deployment-aware model design. I am currently extending this research toward drone-telemetry-based wind-speed estimation and sparse wind-field reconstruction, using commercial UAV attitude signals and co-located meteorological-tower measurements to infer local atmospheric flow fields. Published 19 papers, including 8 first-author works and 1 corresponding-author work, with sustained contributions in wind-energy forecasting, physics-informed learning, edge intelligence, and operational energy prediction.

Publications & Research

Authored 15 journal articles (7 as first author, 1 as corresponding author, and 7 as co-author) and 4 conference papers, with 2 additional manuscripts currently under review.

- **Physics-Informed Multi-Gated Convolutional Recurrent Network for Extreme Wind Speed Prediction.** (Applied Energy, 2026, IF:12.2 , JCR Q1).
  - Developed VORTEX, a physics-informed deep learning framework for extreme wind speed prediction that uses pressure-gradient-force discrepancy as a volatility proxy and adapts learning through a Membership-Gated Convolutional Recurrent Module.
  - On the Jiuquan dataset, VORTEX outperformed ConvLSTM for 6-hour wind speed prediction, reducing RMSE from 1.70 to 1.48 m/s and MAE from 1.23 to 1.06 m/s; in Q4, defined as the 25% most volatile wind cases, the model further improved robustness under extreme-volatility conditions.
- **Physics-Informed Ensemble Learning for City-Center Grid Cell Temperature Prediction During Thermal Extremes** (Urban Climate, 2025, IF: 6.4, JCR Q1).
  - Developed PIEL-NET, a physics-informed hierarchical framework for urban temperature forecasting that couples a simplified advection-diffusion transport model with terrain-aware interaction and spatio-temporal residual learning, thereby capturing wind-driven heat transport and mesoscale thermal evolution.
  - Designed a Physics-Terrain Fusion Agent that combines advective context from the city center and eight surrounding directions with a terrain-mediated coefficient model, then refines the forecast through residual ConvLSTM learning focused on hard extreme-regime errors.
- **Physics-Informed Spatio-Temporal Network with Trainable Adaptive Feature Selection for Short-Term Wind Speed Prediction.** (Computers & Electrical Engineering, 2025, IF: 4.9, JCR Q1).
  - Developed PISTNet, a physics-informed spatio-temporal framework for short-term wind-speed prediction that couples adaptive feature selection with an advection-equation-based physical model to better represent wind-field transport and terrain-sensitive flow behavior.
  - Introduced selective physics-constrained training through Adaptive Physics Penalty Loss, improving forecasting accuracy across four sites and supporting wind-power optimization and grid stability.
- **A Shallow Hybrid Model with Dynamic Bayesian Optimisation for Wind Speed Prediction on Memory-Constrained Devices.** (Computers & Electrical Engineering, 2025, IF: 4.9, JCR Q1).
  - Developed a lightweight hybrid model for short-term wind-speed prediction that combines FastICA, adaptive LASSO, parallel LSTM-TCN temporal modeling, and RVFL-based refinement for resource-limited edge hardware.
  - Achieved a compact accuracy-efficiency balance for edge wind-energy applications, reducing parameter count by 38.04% while improving prediction accuracy and demonstrating deployment on Jetson Nano.

Under Review

- **PiDR-Net: A Perception-Informed Deep Recurrent Network for Short-Term Wind Speed Predictions** (Submitted, Applied Soft Computing).
- **Physics-guided wind-speed estimation from multirotor telemetry using centroid-gated calibration and spectrum-aware temporal correction** (Under Review, Journal of Wind Engineering and Industrial Aerodynamics).

Research Experience

- **Postdoctoral Fellow – Drone Telemetry-Based Wind Estimation and Wind-Field Reconstruction** | Guangdong Technion–Israel Institute of Technology (GTIIT), China | March 2026 – October 2026
  - Conducted research on wind-speed estimation from commercial drone telemetry, using attitude signals, derived tilt variables, and co-located meteorological-tower measurements.
  - Developed physics-guided estimation methods that link UAV tilt response to local wind speed and support sparse-sensing-based wind-field reconstruction.
  - Studied practical wind-sensing workflows for urban climate monitoring, boundary-layer observation, and distributed renewable-energy applications.
- **PhD Researcher – Wind Power Forecasting and Mechanics-Informed Modeling** | Central South University | Sept 2020 – Dec 2025
  - Conducted doctoral research on wind-power forecasting and atmospheric-flow behavior for sustainable energy systems.
  - Developed physics-informed spatio-temporal models for short-term wind-speed prediction under multi-horizon forecasting settings.
  - Integrated data-driven learning with physical and mechanical constraints relevant to wind-field evolution, flow consistency, and engineering interpretability.
  - Designed model architectures, adaptive training losses, and feature-selection strategies for robust forecasting across multiple global wind datasets.
  - Contributed to the Key R&D Program of Hunan Province (Project 2020WK2007) in support of wind-energy storage and grid-oriented optimization.
- **AI/ML Consultant – Edge Computing & Computer Vision** | Bond and Built Pvt Ltd, Pakistan | July 2024 – March 2025
  - Led the applied research and development for a real-time footwear analytics system, tackling challenges in large-scale edge deployment and multi-stream data processing.
  - Scaled a computer vision pipeline to process 50,000+ daily inferences across 60+ concurrent streams, delivering actionable market intelligence and demonstrating robust system architecture.
- **Machine Learning Engineer – Computer Vision & AI Deployment** | DLISION, Pakistan | May 2021 – Sept 2022
  - Led computer vision projects focused on semantic segmentation and real-time object detection.
  - Optimized UNet models with convolutional and attention backbones, improving segmentation accuracy by 3% against their baseline architectures through Focal Loss parameter adjustment.
- **Higher Education Research Assistant – AI for Healthcare** | International Islamic University, Pakistan | March 2019 – Feb 2020
  - Applied Generative Adversarial Networks (GANs) for breast cancer detection, improving classification accuracy.
  - Conducted data augmentation techniques that increased dataset diversity and enhanced model robustness.
- **Lecturer & Lab Engineer – Embedded Systems & AI** | iUSE School of Engineering and Management Sciences, Pakistan | Sept 2013 – Feb 2019
  - Taught undergraduate courses in Embedded Systems, Programming, and Discrete Signal Processing.

- Designed and developed an Instrumentation & Measurement Lab for sensor-based data acquisition in MATLAB.
- Led student research projects on energy management, accident alerts, and secure electronic voting.

Education

- **PhD in Control Science & Engineering** | Central South University, China | Sept 2020 – Dec 2025.
- **MS in Electronic Engineering** | International Islamic University, Pakistan | Sept 2015 – Aug 2017.
- **BS in Electronic Engineering** | International Islamic University, Pakistan | Sept 2006 – Aug 2010.

Technical Skills

- **Research Keywords**: Wind Speed Prediction | Renewable Energy Forecasting | Physics-Informed Machine Learning | Spatio-Temporal Modeling | Time-Series Forecasting | Edge AI.
- **Modeling Expertise**: LSTM/GRU/TCN Architectures | Hybrid Forecasting Models | Adaptive Feature Selection | Multi-Horizon Prediction | Physics-Guided Loss Design.
- **Programming & Tools**: Python | MATLAB | C++ | TensorFlow | PyTorch | OpenCV | Edge Impulse.
- **Deployment & Engineering**: Docker | AWS (EC2) | Git | Triton Inference Server | Edge Inference Evaluation.
- **Hardware Platforms**: NVIDIA Jetson | Raspberry Pi | Arduino Nano BLE Sense.
- **Data Analysis & Optimization**: Pandas | NumPy | Matplotlib | Seaborn | Feature Engineering | Hyperparameter Optimization.

Awards & Honors

- Gold Medalist – MS in Electronic Engineering, International Islamic University.
- Hunan Provincial Government Funding for Research & Development (Project 2020WK2007).
- Chinese Government CSC Scholarship for PhD Studies.

Professional Memberships & Certifications

- Registered Engineer, Pakistan Engineering Council (ELECTRO/22837).
- Member, Institute of Electrical and Electronics Engineers (IEEE).

References

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